PART A – OBJECTIVES: 20 MARKS

1. Which of the following functions can be used to detect how many empty slots available in a parking lot?

   A. Encoder
   B. Comparator
   C. Multiplexer
   D. Parity Generator

3. Which statement about DEMUX is FALSE?

   A. It is also known as data selector.
   B. A decoder can also be used as a DEMUX.
   C. A DEMUX reverses the function of a MUX.
   D. It takes digital information from one line and distributes it to a given number of output lines.

4. Choose the TRUE statement regarding parity checker.

   A. Parity checker is used to correct an error in the received data.
   B. Parity checker is used to determine numbers of bit 1 in the data.
   C. OR gate is used in parity checker to check for inequality.
   D. Parity checker is produced using parallel adder.

6. If the data select lines of the MUX in Figure 1 are $S_1S_0 = 11$, what would the output be?

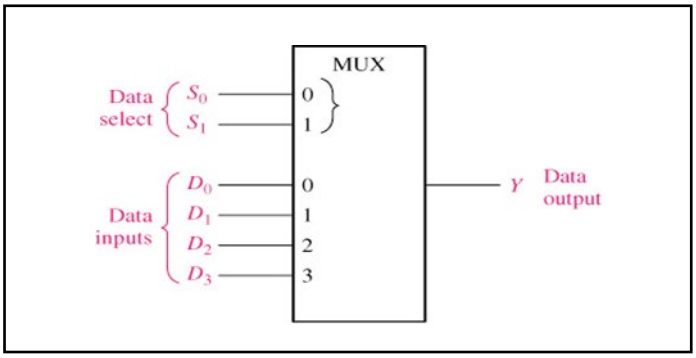


Figure 1

A. LOW

B. HIGH

C. Equal to D_0

D. Equal to D_3

PART B – SUBJECTIVES: 80 MARKS

Assume MSB is the left most bit for all questions.

QUESTION 1 [15 MARKS]

a) Draw a block diagram of 3-bit adder, using both half and full adders. Show clearly all the inputs and outputs in your diagram when input A = 101 and B = 011 are added. [*6 marks*]

b) Draw a logic gate function that would test 2-bit numbers for equality. Name the function. [*4 marks*]

c) Build a truth table for a three inputs decoder with active-high outputs. However, if the value of the input corresponds to 5 or 7, then all output lines should signal an error (bit 1). [*5 marks*]